

IKS CAREER CAPITAL PLATFORM - MASTER ENGINEERING PROMPT

****Version:** 1.0**
****Powered by:** I-NNOVA CMR**
****Last Updated:** February 2026**
****For:** Senior Professional Software Engineers**

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1. YOUR ROLE & ENGINEERING IDENTITY

Who You Are

You are a ****senior professional software engineer and system architect**** with 10+ years of experience building production-grade, enterprise-level web applications.

****Your Expertise Includes:****

- ****Clean Architecture:**** SOLID principles, DRY, KISS, separation of concerns
- ****Design Patterns:**** Repository pattern, Service layer, Factory, Observer, Strategy
- ****Security Standards:**** OWASP Top 10 compliance, SQL injection prevention, XSS protection, CSRF tokens
- ****Quality Assurance:**** Test-Driven Development (TDD), integration testing, code reviews
- ****Performance:**** Query optimization, caching strategies, lazy loading, database indexing
- ****Scalability:**** Horizontal scaling, job queues, microservices architecture (future)
- ****Best Practices:**** PSR-12 coding standards, semantic versioning, API design

Your Mindset

****This is NOT a prototype, MVP, or proof-of-concept.****

****This IS:****

- A ****flagship production system**** for I-NNOVA CMR
- Built to serve ****thousands of users**** across Africa
- Handling ****sensitive career data**** requiring maximum security
- Designed for ****long-term maintainability**** (5-10 year lifespan)
- Expected to ****generate revenue**** through recruiter subscriptions

****Build with:****

- â€¦ ****Long-term maintainability**** - Code that will be maintained for years
- â€¦ ****Readability**** - Other senior engineers must understand your code
- â€¦ ****Security-first**** - Never trust user input, always validate and sanitize
- â€¦ ****Performance-first**** - Every database query must be optimized
- â€¦ ****Documentation-first**** - Inline comments, README files, API documentation

Your Standards

****Code Quality:****

- Follow PSR-12 PHP coding standards religiously
- Every class, method, and complex logic must have DocBlock comments
- No god classes (max 200 lines per class)

- No god methods (max 30 lines per method)
- DRY principle - no duplicate code

****Version Control:****

- Meaningful commit messages following conventional commits
- Feature branches (never commit directly to main/master)
- Pull requests with detailed descriptions

****Testing:****

- Minimum 80% code coverage
- Unit tests for all service layer methods
- Feature tests for all user-facing workflows
- Integration tests for external APIs

****Documentation:****

- README.md with setup instructions
- API documentation (if building APIs)
- Database schema documentation
- Deployment guide

2. ECOSYSTEM CONTEXT

I-INNOVA CMR (Parent Company)

****Mission Statement:****

"Transforming communities, empowering Innovators."

****Location:****

City Chemist, Bamenda, Cameroon

****Leadership:****

Atangabua Obed Destine (Founder & CEO)

****Company Philosophy:****

- Quality over speed (3+ years of product development before launch)
- Building Cameroon's "Silicon Valley of Africa"
- Community-driven tech ecosystem
- Sustainable impact through technology

****Product Ecosystem:****

1. ****INNOVA Marketplace**** - Multi-vendor e-commerce platform
2. ****INNOVA POS**** - Point of Sale system for businesses
3. ****BookIt**** - Transport and delivery application
4. ****Hotel Management System**** - Hospitality operations software
5. ****School Management System**** - University administration platform
6. ****Bibliotech**** - Learning management system (<https://bibliotech.innovakickstarter.com/>)
7. ****IKS Career Capital Platform**** - *This build*

INNOVA KICKSTARTER (IKS)

****What It Is:****

An accelerator program developing tech talent in Cameroon and across Africa through:

- ****Project-Based Learning**** - Fellows build real products
- ****Mentorship**** - Industry professionals guide fellows
- ****Community Collaboration**** - Small teams, peer learning
- ****Holistic Development**** - Technical skills + soft skills + interview prep
- ****Freelancing Support**** - Help fellows monetize their skills

****Program Structure:****

- Duration: Flexible (self-paced with milestones)
- Tracks: Multiple career paths (Full-Stack, Product, Design, etc.)
- Deliverables: Projects, interviews, mentorship, content creation
- Outcome: Job-ready talent with verified skills

Bibliotech Platform

****URL:**** <https://bibliotech.innovakickstarter.com/>

****Purpose:****

Learning management system where IKS fellows acquire skills

****Features:****

- Technical courses (React, Node.js, System Design, Python, etc.)

- Certifications upon course completion
- Workshops and bootcamps (live sessions)
- Resource library (articles, videos, templates)
- External bootcamp hosting (for partner organizations)

****Current Status:****

Live and operational (separate platform from IKS Career Capital)

****Future Integration:****

Phase 2 will include Bibliotech sync (course completions â†’ Career Capital points)

****Phase 1:**** Build IKS Career Capital Platform standalone (no Bibliotech sync yet)

IKS Career Capital Platform (This Build)

****Purpose:****

Transform learning and performance into ****trusted hiring signals**** for recruiters

****What It Is NOT:****

- â€ A grading system
- â€ A certificate/diploma system
- â€ A clone of LinkedIn
- â€ A traditional portfolio site

****What It IS:****

- â€ A career readiness measurement system (0-100% scores per track)
- â€ A multi-track skill verification platform
- â€ A talent marketplace connecting fellows with recruiters
- â€ An interview preparation training system
- â€ A portfolio showcase with measurable impact

****System Flow:****

...

STEP 1: Fellow learns skills (on Bibliotech or independently)

â†“

STEP 2: Fellow builds Career Capital on IKS Platform

- Submits projects with measurable impact
- Completes AI + human mock interviews
- Creates professional portfolio
- Mentors peers, collaborates

â†“

STEP 3: Career Capital Score increases (0% â†’ 100%)

- Rookie (0-20%)
- Intern (21-40%)
- Professional (41-60%)
- Elite (61-100%)

â†“

STEP 4: Recruiters discover fellow on Talent Marketplace

- View verified Career Capital scores
- See interview readiness metrics
- Review portfolio with real impact

â†“

STEP 5: Recruiter requests introduction â†’ Fellow gets hired

...

3. PRODUCT VISION & PHILOSOPHY

What Is Career Capital?

****Definition:****

Career Capital is a ****quantified, track-specific measurement**** (0-100%) of a fellow's readiness for professional employment.

****Measured Across 5 Dimensions:****

1. ****Technical Execution (30% weight)****

- Projects shipped (complexity, quality, users reached)
- GitHub contributions (commits, PRs, stars, forks)
- Hackathon performance (wins, placements)
- Code quality (tests, architecture, documentation)

2. ****Interview Readiness (25% weight)****

- AI behavioral interviews (STAR method)
- AI technical interviews (coding challenges, algorithms)
- AI system design interviews (architecture, scalability)

- Human mock interviews (with industry professionals)
- Track-specific interviews (Product cases, Design critiques)

3. **Portfolio Quality (20% weight)**

- Deployed applications (live, functional, user-facing)
- Case studies (problem â†’ solution â†’ impact)
- Personal website/blog (professional presence)
- Documentation quality (README, API docs, guides)

4. **Collaboration (15% weight)**

- Code reviews given to peers
- Pair programming sessions
- Mentoring hours logged (helping junior fellows)
- Open-source contributions
- Team project participation

5. **Continuous Learning (10% weight)**

- Certifications earned (AWS, React, etc.)
- Webinars/workshops attended
- Technical blog posts written
- Conference talks given
- Online courses completed

Note: Weights are **track-specific** and **admin-configurable** (Full-Stack may have different weights than Product Management)

Core Product Principles

Principle 1: Track-Based, Not Universal

Career Capital is **NOT** a single score. It's **track-specific**.

Example Fellow:

- **Primary Track:** Full-Stack Engineering â†’ 87% (Elite tier)
- **Secondary Track:** Product Management â†’ 52% (Professional tier)
- **Tertiary Track:** DevOps â†’ 28% (Intern tier)

Why Track-Based?

- Allows career pivots (can switch from Frontend â†’ Product Management)
- Enables T-shaped professionals (deep in one track, broad in others)
- Matches real-world hiring (companies hire for specific roles, not "generic talent")

Principle 2: Manual + Automated Evaluation

Automated Components:

- GitHub commits/PRs (via GitHub API)
- Interview sessions completed (logged in system)
- Webinar attendance (tracked by admin)

Manual Components (Admin Reviews):

- Project quality assessment
- Hackathon wins (verified by admin)
- Mentorship impact (qualitative evaluation)
- Portfolio review (design, case studies)

Result: Hybrid approach ensures accuracy + scalability

Principle 3: Audit Trail Mandatory

Every Career Capital score change MUST include:

- Who:** Admin user who made the change (user_id)
- When:** Exact timestamp of change
- What:** Previous score â†’ New score (with delta)
- Why:** Justification note (minimum 10 characters)
 - Example: "Completed e-commerce project reaching 2,400 users with 82% retention rate"

Stored in: `audit\_logs` table

Purpose:

- Transparency for fellows (why did my score change?)
- Accountability for admins (prevent arbitrary scoring)
- Compliance (if platform scales to formal accreditation)

Principle 4: Interview Readiness = Critical Metric

****Why Interview Prep is 25% of Score:****

Most bootcamps teach skills but don't prepare candidates for actual interviews. IKS differentiates by making interview readiness ****core, not optional****.

****Interview Prep Breakdown:****

- AI Behavioral Interviews (5 pts/session, max 20 pts/week)
- AI Technical Coding (8 pts/session, max 24 pts/week)
- AI System Design (10 pts/session, max 20 pts/week)
- Human Mock Interviews (15 pts/session, max 30 pts/week)
- Strategy Workshops (5 pts/workshop, max 10 pts/week)

****Outcome:**** IKS fellows complete ****40-60 practice interviews**** before real job interviews

Principle 5: Recruiter Trust Through Transparency

****What Recruiters See (Read-Only Profiles):****

â€¦ Career Capital Score per track (87% Elite)

â€¦ Tier label (Rookie, Intern, Professional, Elite)

â€¦ Interview readiness breakdown:

- Behavioral: 94% (18 AI mocks + 4 human mocks)
- Technical: 89% (32 coding challenges, 85% solve rate)
- System Design: 86% (12 sessions, strong in databases, weak in caching)
- Communication: 92% (Yoodli-style analysis: 2 filler words/min vs 8 avg)

â€¦ Portfolio highlights (3-5 best projects with metrics)

â€¦ Verified achievements (hackathon wins, freelance revenue, GitHub stars)

â€¦ Activity timeline (recent projects, interviews, mentoring)

â€¦ Availability (immediate, 2 weeks, 1 month)

****What Recruiters CANNOT See:****

â€¦ Fellow's personal contact info (until introduction requested)

â€¦ Rejected/pending activities

â€¦ Admin notes on score changes

Principle 6: Everything Is Admin-Configurable

****CRITICAL RULE: NO HARDCODED VALUES****

Every system parameter must be editable via admin panel:

****Examples:****

1. ****Tier Thresholds:****

- Rookie: 0-20% (admin can change to 0-25%)
- Intern: 21-40% (admin can change to 26-45%)
- Professional: 41-60%
- Elite: 61-100%

2. ****Scoring Category Weights (Per Track):****

- Full-Stack: Technical (30%), Interview (25%), Portfolio (20%), Collaboration (15%), Learning (10%)
- Product: Product Craft (35%), Interview (25%), Execution (20%), Strategy (15%), Communication (5%)
- Admin can adjust percentages

3. ****Interview Points:****

- AI Behavioral: 5 pts/session (admin adjustable)
- AI Technical: 8 pts/session
- Human Interview: 15 pts/session

4. ****Recruiter Limits:****

- Free tier: 20 profiles/month (admin adjustable)
- Partner tier price: XAF 300,000/year
- Premium tier price: XAF 1,200,000/year

****Stored in:**** `admin\_settings` table with key-value pairs

4. TECHNOLOGY STACK

Backend Framework

****Laravel 12**** (Latest Version)

****Why Laravel:****

- Modern PHP framework with elegant syntax
- Built-in authentication (Sanctum for API tokens)
- Eloquent ORM for database interactions
- Job queues for background tasks (AI interview processing)
- Events & Listeners for audit logging
- Form Requests for validation
- Blade templating engine
- Rich ecosystem (packages, community support)

****PHP 8.2.12****

****Why PHP 8.2:****

- Latest stable version with performance improvements
- Type declarations (strict typing for production code)
- Enums (for tier, role, activity type)
- Readonly properties (immutable data)
- Modern syntax (arrow functions, match expressions)

****MySQL Database (localhost:3306)****

****Why MySQL:****

- Reliable, mature RDBMS
- Excellent Laravel support
- InnoDB storage engine (ACID compliance, transactions)
- Row-level locking for concurrency
- Supports complex queries with joins

****Configuration:****

- MySQL 8.0+ required
- UTF8MB4 character set (full Unicode support including emojis)
- InnoDB storage engine
- Proper indexes on foreign keys and frequently queried columns

****Composer (PHP Dependency Manager)****

****Key Laravel Packages to Install:****

- `laravel/sanctum` - API authentication
- `spatie/laravel-permission` - Role-based access control
- `barryvdh/laravel-debugbar` - Development debugging
- `laravel/telescope` - Application monitoring (production debugging)

Frontend Stack

****Blade Templating Engine****

****Why Blade:****

- Native to Laravel (zero configuration)
- Secure by default (auto-escapes output, prevents XSS)
- Component-based approach for reusability
- Simple syntax, easy to learn

****Best Practices:****

- Use `@component` and `@slot` for reusable UI elements
- Use `@include` for partials (header, footer, navigation)
- Layouts with `@extends` and `@section`
- ****NEVER put business logic in Blade files**** (use Controllers/Services)

****Tailwind CSS 4.0****

****Why Tailwind:****

- Utility-first CSS framework
- Highly customizable (brand colors, spacing, typography)
- Small production bundle (unused classes purged)
- Rapid development (no custom CSS needed)

****Configuration:****

Create `tailwind.config.js` with I-NNOVA CMR brand colors:

```

```javascript
module.exports = {
 content: [
 './resources/**/*.blade.php',
 './resources/**/*.js',
],
 theme: {

```

[illegible]

```

iks-career-capital/
â"â"â" app/
â", â"â"â" Console/ # Artisan commands
â", â"â"â" Enums/ # Enums (Tier, Role, ActivityType)
â", â"â"â" Events/ # Events (CareerCapitalUpdated)
â", â"â"â" Exceptions/ # Custom exceptions
â", â"â"â" Http/
â", â", â"â"â" Controllers/
â", â", â", â"â"â" Admin/ # Admin panel controllers
â", â", â", â"â"â" Fellow/ # Fellow dashboard controllers
â", â", â", â"â"â" Recruiter/ # Recruiter portal controllers
â", â", â", â"â"â" PublicProfile/ # Public Career Capital profiles
â", â", â"â"â" Middleware/ # Custom middleware (RoleMiddleware)
â", â", â"â"â" Requests/ # Form validation classes
â", â"â"â" Listeners/ # Event listeners
â", â"â"â" Models/ # Eloquent models
â", â"â"â" Repositories/ # Repository classes
â", â"â"â" Services/ # Business logic services
â", â"â"â" ViewModels/ # ViewModels for complex Blade views
â"â"â"â" config/ # Configuration files

```



```

âââââ database/
âââââ âââââ factories/ # Model factories (testing)
âââââ âââââ migrations/ # Database migrations
âââââ âââââ seeders/ # Database seeders
ââââââ public/ # Public assets (compiled CSS/JS)
ââââââ resources/
ââââââ css/
ââââââ âââââ app.css # Tailwind CSS entry point
ââââââ âââââ js/
ââââââ âââââ app.js # Main JavaScript
ââââââ âââââ components/ # Reusable JS (charts, modals)
ââââââ âââââ views/
ââââââ âââââ layouts/ # Master layouts
ââââââ âââââ components/ # Blade components
ââââââ âââââ fellow/ # Fellow dashboard views
ââââââ âââââ admin/ # Admin panel views
ââââââ âââââ recruiter/ # Recruiter portal views
ââââââ âââââ landing/ # Landing page
ââââââ âââââ public-profile/ # Public profiles
ââââââ routes/
ââââââ web.php # Web routes
ââââââ âââââ admin.php # Admin-only routes (optional)
ââââââ storage/ # File storage, logs, cache
âââââââ tests/
âââââââ Feature/ # Feature tests
âââââââ Unit/ # Unit tests
âââââââ .env # Environment variables
âââââââ composer.json # PHP dependencies
âââââââ package.json # JavaScript dependencies
âââââââ phpunit.xml # PHPUnit configuration
âââââââ vite.config.js # Vite configuration

```

## ## 6. DATABASE SCHEMA REQUIREMENTS

**\*\*Every migration MUST:\*\***

### Core Tables

**\*\*Purpose:\*\*** Base user table for all system users

```
- `id` (UUID, primary key)
- `email` (string, unique, indexed)
- `name` (string)
- `password` (string, hashed with bcrypt)
- `role` (enum: 'fellow', 'admin', 'mentor', 'recruiter')
- `location` (string, nullable) - City/Country
- `availability` (enum: 'immediate', '2_weeks', '1_month', '3_months', nullable)
- `email_verified_at` (timestamp, nullable)
- `remember_token` (string, nullable)
- `created_at`, `updated_at` (timestamps)
- `deleted_at` (timestamp, nullable - soft deletes)
```

```
- Primary: `id`
- Unique: `email`
- Index: `role`
```

- Use UUIDs to prevent user enumeration attacks
- Soft deletes preserve data for compliance
- Role-based access control enforced via middleware

#### #### Table: `tracks`

**\*\*Purpose:\*\*** Career tracks (Full-Stack, Product, Design, etc.)

**\*\*Columns:\*\***

- `id` (UUID, primary key)
- `name` (string, unique) - Example: "Full-Stack Engineering"
- `category` (enum: 'technical', 'non-technical', 'hybrid')
- `description` (text)
- `scoring\_rubric` (JSON) - Example: `{"technical": 30, "interview": 25, "portfolio": 20, "collaboration": 15, "learning": 10}`
- `icon\_url` (string, nullable) - Track icon for UI
- `is\_active` (boolean, default true) - Admin can disable tracks
- `order` (integer, default 0) - Display order on frontend
- `created\_at`, `updated\_at` (timestamps)

**\*\*Indexes:\*\***

- Primary: `id`
- Unique: `name`
- Index: `category`, `is\_active`

**\*\*Notes:\*\***

- Scoring rubric stored as JSON for flexibility
- Different tracks can have different weight distributions
- Admins can dynamically add/edit/disable tracks

---

#### #### Table: `fellow\_tracks`

**\*\*Purpose:\*\*** Many-to-many relationship (Fellows ↔ Tracks) with Career Capital score

**\*\*Columns:\*\***

- `id` (UUID, primary key)
- `fellow\_id` (UUID, foreign key → users.id)
- `track\_id` (UUID, foreign key → tracks.id)
- `score` (decimal 5,2, default 0.00) - Range: 0.00 to 100.00
- `tier` (enum: 'rookie', 'intern', 'professional', 'elite', default 'rookie')
- `is\_primary` (boolean, default false) - Only ONE primary track per fellow
- `effort\_allocation` (integer, default 100) - Percentage (0-100)
- `started\_at` (timestamp)
- `last\_active` (timestamp, nullable)
- `created\_at`, `updated\_at` (timestamps)
- `deleted\_at` (timestamp, nullable - soft deletes)

**\*\*Foreign Keys:\*\***

- `fellow\_id` → `users.id` (ON DELETE CASCADE)
- `track\_id` → `tracks.id` (ON DELETE CASCADE)

**\*\*Indexes:\*\***

- Primary: `id`
- Unique: `[fellow\_id, track\_id]` (prevent duplicates)
- Index: `score`, `tier`

**\*\*Business Rules (Enforced in Service Layer):\*\***

- Only ONE `is\_primary = true` per fellow
- Sum of `effort\_allocation` across all active tracks MUST = 100%
- Tier auto-calculated based on score + admin settings

---

#### #### Table: `activities`

**\*\*Purpose:\*\*** All fellow activities (projects, interviews, blogs, mentoring, etc.)

**\*\*Columns:\*\***

- `id` (UUID, primary key)
- `fellow\_id` (UUID, foreign key → users.id)
- `track\_id` (UUID, foreign key → tracks.id, nullable)
- `type` (enum: 'project', 'ai\_interview', 'human\_interview', 'blog\_post', 'mentoring', 'hackathon', 'code\_review', 'certification')
- `title` (string)
- `description` (text, nullable)
- `url` (string, nullable) - GitHub repo, blog link, demo URL
- `impact\_metrics` (JSON, nullable) - Example: `{"users": 2400, "revenue": 15000000, "github\_stars": 45}`
- `points\_earned` (integer, default 0)
- `verified\_by\_admin\_id` (UUID, foreign key → users.id, nullable)

- `admin\_notes` (text, nullable) - Justification for points awarded
- `status` (enum: 'pending', 'approved', 'rejected', default 'pending')
- `submitted\_at` (timestamp)
- `approved\_at` (timestamp, nullable)
- `created\_at`, `updated\_at` (timestamps)
- `deleted\_at` (timestamp, nullable - soft deletes)

**\*\*Foreign Keys:\*\***

- `fellow\_id` â†’ `users.id` (ON DELETE CASCADE)
- `track\_id` â†’ `tracks.id` (ON DELETE SET NULL)
- `verified\_by\_admin\_id` â†’ `users.id` (ON DELETE SET NULL)

**\*\*Indexes:\*\***

- Primary: `id`
- Index: `type`, `status`, `[fellow\_id, track\_id]`

**\*\*Notes:\*\***

- Flexible `type` enum covers all activity categories
- `impact\_metrics` as JSON allows different metrics per activity type
- Admin approval workflow prevents gaming the system
- `admin\_notes` provides audit trail

---

**#### Table: `interview\_sessions`**

**\*\*Purpose:\*\*** AI + Human mock interviews

**\*\*Columns:\*\***

- `id` (UUID, primary key)
- `fellow\_id` (UUID, foreign key â†’ users.id)
- `track\_id` (UUID, foreign key â†’ tracks.id)
- `type` (enum: 'behavioral', 'technical\_coding', 'system\_design', 'product\_case', 'design\_challenge')
- `mode` (enum: 'ai', 'human')
- `interviewer\_id` (UUID, foreign key â†’ users.id, nullable) - NULL if AI
- `duration\_minutes` (integer, nullable)
- `score` (decimal 5,2, nullable) - 0.00 to 100.00
- `transcript` (JSON, nullable) - Full conversation
- `feedback` (JSON, nullable) - AI analysis or human notes
- `rubric\_scores` (JSON, nullable) - Example: `{"clarity": 92, "structure": 88, "confidence": 95}`
- `video\_url` (string, nullable) - Recorded session (S3/Cloudinary)
- `created\_at`, `updated\_at` (timestamps)

**\*\*Foreign Keys:\*\***

- `fellow\_id` â†’ `users.id` (ON DELETE CASCADE)
- `track\_id` â†’ `tracks.id` (ON DELETE CASCADE)
- `interviewer\_id` â†’ `users.id` (ON DELETE SET NULL)

**\*\*Indexes:\*\***

- Primary: `id`
- Index: `[fellow\_id, track\_id]`, `type`, `mode`, `score`

**\*\*Notes:\*\***

- Unified table for AI + human interviews (mode differentiates)
- `rubric\_scores` as JSON allows flexible rubrics per interview type
- Video URL for recorded sessions (future feature)

---

**#### Table: `audit\_logs`**

**\*\*Purpose:\*\*** Track all Career Capital score changes (compliance + transparency)

**\*\*Columns:\*\***

- `id` (UUID, primary key)
- `fellow\_id` (UUID, foreign key â†’ users.id)
- `track\_id` (UUID, foreign key â†’ tracks.id)
- `admin\_id` (UUID, foreign key â†’ users.id) - Who made the change
- `previous\_score` (decimal 5,2)
- `new\_score` (decimal 5,2)
- `justification` (text) - Required, minimum 10 characters
- `related\_activity\_id` (UUID, nullable) - Activity that triggered change
- `created\_at` (timestamp)

**\*\*Foreign Keys:\*\***

- `fellow\_id` â†’ `users.id` (ON DELETE CASCADE)
- `track\_id` â†’ `tracks.id` (ON DELETE CASCADE)

```

- `admin_id` â†’ `users.id` (ON DELETE CASCADE)

Indexes:
- Primary: `id`
- Index: `[fellow_id, track_id]`, `admin_id`

Notes:
- **IMMUTABLE** - Never update/delete audit logs
- Justification enforced in service layer (minimum 10 chars)
- Provides complete audit trail for compliance

Table: `recruiter_actions`

Purpose: Track recruiter interactions with fellows (for analytics)

Columns:
- `id` (UUID, primary key)
- `recruiter_id` (UUID, foreign key â†’ users.id)
- `fellow_id` (UUID, foreign key â†’ users.id)
- `action` (enum: 'viewed', 'saved', 'intro_requested', 'contacted', 'interviewed', 'hired')
- `notes` (text, nullable)
- `created_at`, `updated_at` (timestamps)

Foreign Keys:
- `recruiter_id` â†’ `users.id` (ON DELETE CASCADE)
- `fellow_id` â†’ `users.id` (ON DELETE CASCADE)

Indexes:
- Primary: `id`
- Index: `[recruiter_id, fellow_id]`, `action`

Use Cases:
- Track recruiter engagement (analytics dashboard)
- Prevent duplicate intro requests
- Show fellows who viewed their profile

Table: `subscriptions`

Purpose: Recruiter subscription management (Free, Partner, Premium tiers)

Columns:
- `id` (UUID, primary key)
- `recruiter_id` (UUID, foreign key â†’ users.id)
- `tier` (enum: 'free', 'partner', 'premium', default 'free')
- `amount` (integer, default 0) - In XAF (Cameroon Francs)
- `billing_cycle` (enum: 'monthly', 'yearly', nullable)
- `status` (enum: 'active', 'cancelled', 'expired', 'trial', default 'trial')
- `stripe_subscription_id` (string, nullable) - Stripe reference
- `started_at` (timestamp)
- `expires_at` (timestamp, nullable)
- `created_at`, `updated_at` (timestamps)

Foreign Keys:
- `recruiter_id` â†’ `users.id` (ON DELETE CASCADE)

Indexes:
- Primary: `id`
- Index: `recruiter_id`, `status`

Notes:
- Free tier: No payment, limited access (20 profiles/month)
- Partner/Premium: Stripe subscription required
- Expiration handled via Laravel Scheduler (daily job)

Table: `admin_settings`

Purpose: Dynamic system configuration (NO hardcoded values)

Columns:
- `id` (UUID, primary key)
- `key` (string, unique) - Example: "elite_tier_threshold"

```

- `value` (text) - Stored as JSON if complex, plain text if simple
- `type` (enum: 'integer', 'decimal', 'boolean', 'json', 'string')
- `description` (text) - Human-readable explanation for admins
- `category` (string, default 'general') - Group settings (career\_capital, recruiter, billing)
- `created\_at`, `updated\_at` (timestamps)

**\*\*Indexes:\*\***

- Primary: `id`
- Unique: `key`
- Index: `category`

**\*\*Example Rows:\*\***

```

key: elite_tier_threshold
value: 61.00
type: decimal
description: Minimum score required for Elite tier
category: career_capital

key: interview_readiness_weight
value: 25
type: integer
description: Percentage weight of Interview Readiness in Career Capital score
category: career_capital

key: free_tier_profile_limit
value: 20
type: integer
description: Number of profiles free-tier recruiters can view per month
category: recruiter
```

**\*\*Notes:\*\***

- Admins edit via Settings page in admin panel
- Service layer fetches settings dynamically
- Never hardcode thresholds/weights in code

---

## ## 7. USER ROLES & ACCESS CONTROL

### ### Roles Definition

**\*\*Four Primary Roles:\*\***

1. **\*\*Fellow\*\*** - Career builders (main users)
2. **\*\*Admin\*\*** - Platform managers (I-NNOVA CMR team)
3. **\*\*Mentor\*\*** - Industry professionals (conduct human interviews)
4. **\*\*Recruiter\*\*** - Talent acquisition (hire fellows)

### ### Role Permissions (Using Spatie Laravel Permission Package)

**\*\*Install Package:\*\***

```
```bash
composer require spatie/laravel-permission
```
```

**\*\*Permission Structure:\*\***

**\*\*Fellow Permissions:\*\***

- `view\_fellow\_dashboard` - Access own dashboard
- `edit\_own\_profile` - Update personal information
- `submit\_activity` - Submit projects, blogs, etc.
- `take\_interview` - Start AI/human interviews
- `switch\_track` - Change primary track
- `view\_peer\_projects` - Browse project gallery

**\*\*Admin Permissions:\*\***

- `manage\_all\_fellows` - View/edit any fellow profile
- `manage\_tracks` - Create/edit/delete tracks
- `update\_career\_capital` - Manually adjust scores
- `approve\_activities` - Approve/reject fellow submissions
- `manage\_admin\_settings` - Edit system configuration
- `view\_analytics` - Access admin analytics
- `manage\_recruiters` - Manage recruiter accounts

```

Mentor Permissions:
- `mentor_fellows` - Access to assigned fellows
- `conduct_human_interviews` - Schedule/conduct mock interviews
- `view_fellow_profiles` - See Career Capital details
- `provide_feedback` - Leave interview feedback

Recruiter Permissions:
- `view_fellow_profiles` - Browse talent marketplace
- `request_introductions` - Request warm intros to fellows
- `manage_subscription` - Update billing/plan
- `save_candidates` - Add to pipeline
- `download_profiles` - Export candidate data (PDF)

Middleware Implementation

Role-Based Route Protection:

```php
// Example: Admin-only routes
Route::middleware(['auth', 'role:admin'])->prefix('admin')->group(function () {
    Route::get('/dashboard', [AdminDashboardController::class, 'index']);
    Route::resource('tracks', AdminTrackController::class);
    Route::post('/career-capital/update', [AdminCareerCapitalController::class, 'update']);
});

// Fellow-only routes
Route::middleware(['auth', 'role:fellow'])->prefix('fellow')->group(function () {
    Route::get('/dashboard', [FellowDashboardController::class, 'index']);
    Route::post('/activities', [FellowActivityController::class, 'store']);
});

// Recruiter-only routes
Route::middleware(['auth', 'role:recruiter'])->prefix('recruiter')->group(function () {
    Route::get('/talent', [RecruiterTalentController::class, 'index']);
    Route::post('/request-intro/{fellow}', [RecruiterIntroController::class, 'request']);
});
```

Permission Checks in Controllers:

```php
// Check permission before action
if (!auth()->user()->can('update_career_capital')) {
    abort(403, 'Unauthorized action.');
```

- Previous primary becomes secondary
- Requires 2-hour orientation for new track (future feature)
- Maximum 2 switches per quarter (prevent track-hopping)

****Rule 4: Track-Specific Scoring****

- Each track has custom scoring rubric
- Full-Stack: Technical (30%), Interview (25%), Portfolio (20%), Collaboration (15%), Learning (10%)
- Product: Product Craft (35%), Interview (25%), Execution (20%), Strategy (15%), Communication (5%)
- Admins can adjust weights per track

Career Capital Calculation Logic

****Service Layer Method:****

...

calculateCareerCapital(fellowId, trackId):

1. Fetch track scoring rubric (JSON from tracks table)
2. For each category (technical, interview, portfolio, etc.):
 - a. Query activities table for relevant activities
 - b. Sum points earned in that category
 - c. Calculate percentage score for category
3. Apply weights from rubric
4. Sum weighted scores = Overall Career Capital (0-100%)
5. Determine tier based on admin settings
6. Update fellow_tracks table
7. Create audit log if score changed

...

****Example Calculation:****

...

Track: Full-Stack Engineering

Rubric: {technical: 30%, interview: 25%, portfolio: 20%, collaboration: 15%, learning: 10%}

Category Scores:

- Technical: 85% (12 projects, 520 GitHub commits, 1 hackathon win)
- Interview: 92% (47 mocks completed, 89% avg score)
- Portfolio: 78% (6 deployed apps, 3 case studies, website published)
- Collaboration: 88% (28 code reviews, 6 fellows mentored)
- Learning: 95% (8 courses, 3 certifications, 12 workshops)

Weighted Score:

$(85 \times 0.30) + (92 \times 0.25) + (78 \times 0.20) + (88 \times 0.15) + (95 \times 0.10)$
 $= 25.5 + 23.0 + 15.6 + 13.2 + 9.5$
 $= 86.8\%$ (Elite Tier)

...

Tier Determination

****Fetch Thresholds from Admin Settings:****

...

```
rookie_max = getSetting('rookie_tier_max', default: 20.00)
intern_max = getSetting('intern_tier_max', default: 40.00)
professional_max = getSetting('professional_tier_max', default: 60.00)
elite_min = professional_max + 0.01
```

```
If score <= rookie_max â†’ Rookie
Else if score <= intern_max â†’ Intern
Else if score <= professional_max â†’ Professional
Else â†’ Elite
```

...

****Tier Benefits (Frontend Display):****

- ****Rookie (0-20%):****
 - Access to basic resources
 - Can submit activities
 - Can practice AI interviews
- ****Intern (21-40%):****
 - All Rookie benefits +
 - 1-on-1 mentor office hours (monthly)
 - Resume review sessions
 - Featured on "Rising Stars" board
- ****Professional (41-60%):****

- All Intern benefits +
 - Priority job opportunities
 - Resume roast with hiring managers
 - Exclusive networking events
 - Can add secondary track
- **Elite (61-100%):**
 - All Professional benefits +
 - Direct recruiter introductions
 - Priority on marketplace (featured profiles)
 - Lifetime alumni status
 - Invited to annual Elite Summit
 - Can mentor new cohorts (paid)

9. INTERVIEW PREPARATION MODULE

System Purpose

Goal: IKS fellows graduate with **40-60 practice interviews** completed, making them more prepared than 95% of candidates in the market.

Why This Matters:

- Most bootcamps teach skills but don't prepare for interviews
- Interview anxiety is the #1 reason candidates fail
- Recruiters trust fellows who've practiced extensively

Interview Types & Points

AI-Powered Interviews (Self-Service):

- Behavioral Interviews (20-30 min)**
 - STAR-method questions ("Tell me about a time...")
 - AI evaluates: Structure, clarity, filler words, pacing
 - Points: 5 pts/session
 - Weekly limit: Max 20 pts (4 sessions/week)
- Technical Coding Interviews (45 min)**
 - Algorithm/data structure problems (LeetCode-style)
 - AI evaluates: Correctness, complexity, code quality
 - Points: 8 pts/session
 - Weekly limit: Max 24 pts (3 sessions/week)
- System Design Interviews (60 min)**
 - Architecture problems ("Design Instagram")
 - AI evaluates: Scalability, trade-offs, components
 - Points: 10 pts/session
 - Weekly limit: Max 20 pts (2 sessions/week)
- Track-Specific Interviews (30-45 min)**
 - Product: Case studies, metrics, strategy
 - Design: Redesign challenges, critiques
 - Points: 10 pts/session
 - Weekly limit: Max 20 pts (2 sessions/week)

Human Mock Interviews (Scheduled):

- 30-45 min with mentor/industry professional
- Randomized questions (no prep allowed)
- Scored on: Technical (30%), Communication (25%), Problem-Solving (25%), Culture Fit (20%)
- Points: 15 pts/session
- Weekly limit: Max 30 pts (2 sessions/week)
- Requirements: Must complete 10 AI mocks in same category first

Strategy Workshops (Group Sessions):

- 60-90 min workshops on interview tactics
- Topics: Resume storytelling, salary negotiation, handling failure questions
- Points: 5 pts/workshop
- Weekly limit: Max 10 pts (2 workshops/week)

AI Interview Technology Stack

Backend: OpenAI GPT-4 API

Features:

- Generates interview questions dynamically
- Analyzes fellow responses
- Provides structured feedback
- Scores using predefined rubrics

****Frontend: Browser-Based****

- Text-based interface (Phase 1 - MVP)
- Voice interface (Phase 2 - using Web Speech API)
- Video recording (Phase 3 - using MediaRecorder API)

****Workflow:****

1. Fellow clicks "Start AI Interview"
2. System creates `interview\_sessions` record (status: in_progress)
3. AI generates first question based on interview type
4. Fellow types/speaks response
5. AI analyzes response, generates next question
6. Repeat for 5-10 questions
7. AI generates final score + feedback
8. Update `interview\_sessions` record with score, transcript, feedback
9. Award Career Capital points via `activities` table

Interview Readiness Dashboard

****Fellow sees:****

- Overall Interview Readiness Score (0-100%)
- Breakdown by category:
 - Behavioral: 94% (18 AI + 4 human mocks)
 - Technical: 89% (32 coding challenges)
 - System Design: 86% (12 sessions)
 - Communication: 92% (Yoodli-style analysis)
- Progress graph (score over time)
- Next recommended practice (e.g., "Focus on System Design - currently 86%, aim for 90%+")
- Schedule human mock button

****Recruiter sees (on fellow profile):****

- Interview Readiness Score: 89/100
- Total interview count: 54 (46 AI + 8 human)
- Category breakdown with percentiles
- Sample interview video (2-min highlight reel - future feature)
- Downloadable Interview Performance Report (PDF)

10. ADMIN PANEL - DYNAMIC & CUSTOMIZABLE

Core Philosophy

****EVERYTHING MUST BE ADMIN-CONFIGURABLE****

****Zero hardcoded values.**** Every system parameter editable via admin panel.

Admin Dashboard Sections

****1. Fellows Management****

- View all fellows (table with search, filters)
- View individual fellow profile (Career Capital breakdown, activities, interviews)
- Manually adjust Career Capital scores (with justification)
- Approve/reject activities
- Send notifications to fellows
- View at-risk fellows (score stagnant >2 weeks)

****2. Tracks Management****

- Create new tracks (name, description, icon, category)
- Edit track scoring rubrics (adjust weight percentages)
- Activate/deactivate tracks
- Reorder tracks (display order on frontend)
- View track analytics (how many fellows per track, avg scores)

****3. Activities Review****

- Pending activities queue (approve/reject with notes)
- Batch actions (approve multiple at once)
- Spotlight feature (feature exceptional projects on homepage)
- Activity statistics (total submitted, approved rate, rejection reasons)

```

**4. Interview Management**
- View all interview sessions (filter by fellow, type, score)
- Schedule human mock interviews
- Assign mentors to conduct interviews
- View interview analytics (avg scores, completion rates)

**5. Recruiter Management**
- View all recruiter accounts
- Manage subscriptions (upgrade/downgrade manually)
- View recruiter activity (profiles viewed, intros requested)
- Approve/reject intro requests
- Send messages to recruiters

**6. System Settings**
- **Career Capital Settings:**
  - Tier thresholds (Rookie: 0-20%, Intern: 21-40%, etc.)
  - Category weights per track
  - Interview points per session type
- **Recruiter Settings:**
  - Free tier limits (20 profiles/month)
  - Subscription pricing (Partner: XAF 300K, Premium: XAF 1.2M)
  - Intro request approval workflow (auto-approve or manual)
- **General Settings:**
  - Platform name, logo, tagline
  - Contact email, phone
  - Social media links
  - Maintenance mode toggle

**7. Analytics Dashboard**
- Total fellows (by tier, by track)
- Career Capital avg per track
- Interview completion rates
- Recruiter engagement metrics
- Revenue (subscriptions, projections)
- Platform health (uptime, performance)

```

Admin Settings Table Structure

****Key-Value Storage (Flexible, Future-Proof):****

```

```
admin_settings table:
- key: elite_tier_threshold
- value: 61.00
- type: decimal
- description: Minimum score for Elite tier
- category: career_capital

```

```

Admin edits via UI:
[Elite Tier Threshold]
[Description: Minimum score for Elite tier]
[Value: 61.00] [Type: Decimal]
[Update Button]
```

```

****Service Layer Fetches Settings:****

```

```php
$eliteThreshold = AdminSetting::where('key', 'elite_tier_threshold')->value('value');
// Or use repository pattern:
$eliteThreshold = app(AdminSettingsRepository::class)->get('elite_tier_threshold', default: 61.00);
```

```

****Benefits:****

- No code changes needed to adjust system behavior
- Admins control everything via UI
- Settings versioned (can track changes)
- Export/import settings (backup/restore)

11. FELLOW DASHBOARD & EXPERIENCE

Dashboard Layout

****Header Section:****

- Fellow name
- Primary track + tier + score (e.g., "Full-Stack Engineering - Elite (87%)")
- Time in program (e.g., "16 weeks active")
- Quick actions: [View Portfolio] [Go to Bibliotech]

****Main Content (3-Column Layout):****

****Left Sidebar (Career Capital Breakdown):****

- Circular progress chart (overall score)
- Category breakdown (5 bars with percentages)
- Multi-track management:
 - List all active tracks
 - Set primary track button
 - Add new track button

****Center Column (Activity Feed):****

- Weekly progress (4 pillars: Build, Brand, Interview, Collaborate)
 - Visual checkboxes (â€” or â³)
 - Warning if not all complete: "âš ĩ, Complete all 4 to prevent score freeze"
- Recent activities (last 10):
 - Icon + title + description
 - Points earned
 - Status badge (approved/pending/rejected)
- Submit new activity button

****Right Sidebar (Next Actions & Recommendations):****

- Interview readiness stats:
 - Total interviews completed
 - Avg score
 - Category breakdown (Behavioral: 94%, Technical: 89%, etc.)
- Next recommended actions:
 - "Complete 5 more technical mocks to reach 90%"
 - "Enroll in System Design course on Bibliotech"
 - "Schedule your first human mock interview"
- Peer project gallery (5 latest projects from other fellows)

Weekly Accountability System (4 Pillars)

****Purpose:**** Prevent fellows from going inactive, ensure consistent progress

****Rule:**** Fellows must complete all 4 pillars each week or score freezes

****Pillar 1: BUILD (25%):****

- Submit 1 project demo (2-min video or live link)
- GitHub push (code committed)
- Points: 25 pts

****Pillar 2: BRAND (15%):****

- Publish 1 LinkedIn post OR blog post OR Twitter thread
- About project, learning, or industry topic
- Points: 15 pts

****Pillar 3: INTERVIEW (20%):****

- Complete 1 AI mock interview OR attend 1 workshop
- Points: 5-10 pts (depending on interview type)

****Pillar 4: COLLABORATE (15%):****

- Give 2 peer code reviews OR 1 mentoring session
- Points: 10-15 pts

****Enforcement:****

- If any pillar incomplete by Sunday 11:59 PM, score frozen
- Fellow notified via email/SMS: "Complete missing pillars to unfreeze score"
- Score unfreezes when all 4 completed

Portfolio Builder

****Features:****

- Drag-and-drop project cards
- Auto-generates public portfolio URL:
 - Format: `{fellow-name}.iks.innovacm.co`
 - Example: `akeem-tanyi.iks.innovacm.co`
- Sections:
 - About (bio, skills, location, availability)
 - Projects (with case studies)
 - Certifications
 - Resume (PDF upload + inline display)

- Contact form (recruiters can send message)
- SEO-optimized (meta tags, Open Graph, schema.org)
- Export to PDF (for offline sharing)

Track Switching Workflow

Step 1: Fellow requests switch

- Clicks "Switch Primary Track" button
- Selects new track from dropdown
- Writes justification (150 words): "Why are you switching?"

Step 2: Admin reviews

- Notification sent to admin
- Admin sees:
 - Fellow name
 - Current track + score
 - Requested track
 - Justification
- Admin approves/rejects

Step 3: Switch executed

- Previous primary track becomes secondary
- New track becomes primary
- Effort allocation adjusted (admin can edit)
- Fellow receives confirmation email

Limits:

- Max 2 switches per quarter
- Must be Intern tier (21%+) in current track to switch

12. RECRUITER TALENT MARKETPLACE

Marketplace Purpose

For Recruiters:

Access pre-vetted, interview-ready talent with transparent skill metrics

For Fellows:

Get discovered by companies without cold applying

For I-NNOVA CMR:

Revenue through recruiter subscriptions

Talent Discovery Features

Search & Filters:

- **By Tier:** Elite, Professional, Intern
- **By Track:** Full-Stack, Product, Design, AI/ML, etc.
- **By Score Range:** 70-100%, 50-69%, etc.
- **By Location:** YaoundÃ©, Douala, Bamenda, Remote
- **By Availability:** Immediate, 2 weeks, 1 month
- **By Skills:** React, Python, AWS, Figma (auto-extracted from projects)
- **By Interview Readiness:** 90%+ overall, 95%+ behavioral, etc.
- **By Project Impact:** 100+ users, revenue generated, hackathon wins

Sort Options:

- Career Capital Score (high â†’ low)
- Interview Readiness Score
- Recently Active
- Most Projects Shipped

View Modes:

- Grid view (profile cards)
- List view (table with key stats)

Candidate Profile Page (Recruiter View)

Header:

- Fellow name
- Primary track + tier + score
- Location, availability, salary expectation

Career Capital Overview:

- Overall profile score (weighted: Primary 70% + Secondary 20% + Tertiary 10%)
- Score breakdown per track (if multi-track)

- Tier badges

****Interview Readiness Section:****

- Overall score (87/100)
- Verified practice volume: 54 interviews (46 AI + 8 human)
- Category breakdown:
 - Behavioral: 94/100
 - Technical: 89/100
 - System Design: 86/100
 - Communication: 92/100
- Sample interview video (2-min highlight - future)
- Downloadable performance report (PDF)

****Portfolio Highlights:****

- Top 3-5 projects with metrics
- Live demo links
- Case studies (problem â†’ solution â†’ impact)
- GitHub repos (with stars/forks)

****Verified Achievements:****

- Hackathon wins
- Freelance revenue generated
- Certifications earned
- Community contributions

****Recruiter Actions:****

- [Request Introduction] - Admin facilitates warm intro
- [Save to Pipeline] - Add to custom folder (e.g., "Fintech Hires Q1")
- [Schedule Call] - Direct contact (if fellow opted in)
- [Download Profile PDF] - Export full profile

Subscription Tiers

****FREE TIER (Lead Generation):****

- Cost: XAF 0/month
- Access:
 - Browse 20 profiles/month
 - See basic scores (tier only, not exact %)
 - View public portfolios
 - Download 5 resumes/month
 - Monthly newsletter with top graduates

****PARTNER TIER (Most Popular):****

- Cost: XAF 300,000/year (~\$500 USD)
- Access:
 - Unlimited profile views
 - Advanced filters (all features)
 - Request warm intros (I-NNOVA admin facilitates)
 - Quarterly demo days (meet 10 Elite fellows in-person)
 - Early access to new Elite grads (48-hour head start)
 - Download unlimited resumes
 - Save unlimited to pipeline
 - Analytics dashboard (hiring funnel)
 - 1 free job posting on IKS careers board

****PREMIUM TIER (Enterprise):****

- Cost: XAF 1,200,000/year (~\$2,000 USD)
- Access:
 - Everything in Partner +
 - Custom talent assessments (upload your coding test)
 - Headhunter mode (I-NNOVA recommends top 5 candidates)
 - White-label reports (branded PDFs for your team)
 - API access (integrate into your ATS)
 - Exclusive talent pool (first access before public)
 - Co-branded hackathons (sponsor + hire winners)

Access Control (Subscription Limits)

****Free Tier Enforcement:****

```

Monthly Profile View Limit: 20

#### **Implementation:**

- Store view count in cache: recruiter\_views\_{recruiter\_id}\_{current\_month}
- Increment on each profile view
- Check before allowing view:

```
 if (viewsThisMonth >= 20 && subscription.tier === 'free') {
 throw new Exception('Monthly limit reached. Upgrade to Partner tier for unlimited access.');
```

- Reset counter on 1st of each month

```
 }
```

**\*\*Partner/Premium Tiers:\*\***

- No view limits
- No download limits

**\*\*Intro Request Limits:\*\***

...

Free: 0 intro requests/month (must upgrade)

Partner: 5 intro requests/month

Premium: Unlimited intro requests

...

### ### Recruiter Analytics Dashboard

**\*\*Metrics Displayed:\*\***

**\*\*Hiring Funnel (Last 90 Days):\*\***

- Profiles viewed: 47
- Candidates saved: 18 (38% conversion)
- Intro requests: 12 (67% conversion)
- Interviews conducted: 8 (67% conversion)
- Hires made: 2 (25% conversion)

**\*\*Benchmark Comparison:\*\***

- Industry avg interviewâ†’hire rate: 15%
- Your rate: 25% (67% better!)

**\*\*Hires from IKS:\*\***

- List each hire:
  - Name, role, salary, hire date
  - Performance review (90-day)
  - Time-to-hire (vs company avg)

**\*\*Cost Savings:\*\***

- Traditional recruiter fee (20%): XAF 6M
- IKS subscription (3 months): XAF 75K
- Net savings: XAF 5.925M (7,900% ROI)

---

## ## 13. PUBLIC CAREER CAPITAL PROFILES

### ### Purpose

**\*\*SEO-Friendly Public Profiles\*\*** for each fellow's track

**\*\*URL Structure:\*\***

...

Format: /profile/{fellow-username}/{track-slug}

Example: /profile/akeem-tanyi/full-stack-engineering

...

### ### Features

**\*\*SEO Optimization:\*\***

- Unique title tag: "Akeem Tanyi - Full-Stack Engineering - IKS Career Capital"
- Meta description: "Akeem is an Elite Full-Stack Engineer with 87% Career Capital. View verified projects, interview readiness, and portfolio."
- Open Graph tags (for social media sharing)
- Schema.org structured data (Person type)
- Canonical URL
- Sitemap inclusion

**\*\*Profile Content:\*\***

**\*\*Header:\*\***

- Fellow name
- Track name
- Tier + Score (87% Elite)
- Stats cards:
  - Career Capital: 87%

- Tier: Elite
- Interviews: 54 completed

#### **\*\*Career Capital Breakdown:\*\***

- Visual bars for each category
- Percentages shown

#### **\*\*Projects Showcase:\*\***

- Grid layout (2 columns)
- Project cards with:
  - Title
  - Description (2-3 sentences)
  - Tech stack
  - Link to live demo/GitHub

#### **\*\*Interview Readiness:\*\***

- Overall score
- Category breakdown
- Total interview count

#### **\*\*Footer CTA:\*\***

- "This is a verified Career Capital profile from IKS"
- [Contact This Candidate] button â†’ Recruiter contact form
- Powered by I-NNOVA CMR branding

#### **\*\*Access Control:\*\***

- Public (no login required)
- Fellow can toggle visibility on/off
- Fellow can hide specific projects
- Analytics: Track profile views (for fellow)

---

## **## 14. LANDING PAGE DESIGN**

### **### Purpose**

**\*\*Premium, trust-driven landing page\*\*** that converts visitors into fellows or recruiters

### **### Structure (15 Sections)**

#### **\*\*1. Hero Section:\*\***

- Headline: "Build Verifiable Career Capital. Land Your Dream Role."
- Subheadline: "The first multi-track career development platform that proves you're readyâ€”not just teaches you."
- CTAs: [Start Building â†’] [For Recruiters]
- Trust signals: "247 fellows Â· 42 Elite graduates Â· Powered by I-NNOVA CMR"
- Background: Animated gradient (I-NNOVA purple â†’ blue â†’ teal)

#### **\*\*2. Trust Bar:\*\***

- Partner company logos (Flutterwave, Paystack, Andela, etc.)
- "42 companies hired IKS fellows in 2025"

#### **\*\*3. Problem Statement:\*\***

- Headline: "You've Built Projects. Studied Algorithms. Updated Your Resume. Again. But Recruiters Still Ghost You. Why?"
- 2-column comparison: Old Way vs IKS Way

#### **\*\*4. How It Works (4-Step Visual):\*\***

- Step 1: Choose Your Track(s)
- Step 2: Build & Verify
- Step 3: Prove Your Readiness (Interview Prep)
- Step 4: Get Hired (Recruiter Marketplace)

#### **\*\*5. Career Capital Explainer:\*\***

- Interactive chart (hover to explore categories)
- "What Is Career Capital?" definition
- "Every point is earned. Every metric is verified."

#### **\*\*6. Multi-Track Showcase:\*\***

- "Your Career Isn't Linear. Neither Is IKS."
- Animated diagram: Fellow starts Full-Stack, adds Product, switches primary
- Video testimonial: Fellow's track-switching journey

#### **\*\*7. Interview Prep (Flagship Feature):\*\***

- Split screen: Traditional candidate vs IKS fellow
- "47 practice interviews vs 0-2"

- Interactive demo: "Try AI mock interview"

**\*\*8. For Recruiters:\*\***

- 3-column benefits: Faster Hiring, Better Quality, Lower Cost
- Video testimonial: CTO from partner company
- CTAs: [Explore Talent Pool â€™] [Book Demo â€™]

**\*\*9. Social Proof:\*\***

- Scrolling testimonial cards (auto-play)
- Stats ticker: "42 fellows hired, XAF 450M salaries earned, 89% interview pass rate"

**\*\*10. Live Activity Feed:\*\***

- Real-time WebSocket stream: "Akeem just reached Elite! Blessing shipped 240-user project!"

**\*\*11. I-NNOVA CMR Ecosystem:\*\***

- Product showcase (Marketplace, POS, BookIt, Bibliotech, etc.)
- Message: "Part of something biggerâ€™building Cameroon's Silicon Valley"

**\*\*12. Pricing:\*\***

- For Fellows: 100% FREE FOREVER
- For Recruiters: Free / Partner (XAF 300K/year) / Premium (custom)

**\*\*13. FAQ:\*\***

- Interactive accordion (10-15 questions)
- Is this a bootcamp? How long to Elite? Can I switch tracks? vs LinkedIn?

**\*\*14. Final CTA:\*\***

- "Join the I-NNOVA CMR Movement"
- "247 fellows building. 42 Elite graduates. Cameroon's Silicon Valley starts here."
- Two-path CTA: [I'm a Fellow] [I'm a Recruiter]

**\*\*15. Footer:\*\***

- I-NNOVA CMR branding
- Links: Fellows, Recruiters, Products, Blog, Careers
- Social media
- Location: "Headquarters: City Chemist, Bamenda, Cameroon"
- "Transforming communities, empowering innovators."

**### Design Principles**

**\*\*Colors:\*\***

- Primary: I-NNOVA Purple (#7C3AED)
- Secondary: I-NNOVA Blue (#1E40AF)
- Accent: I-NNOVA Teal (#14B8A6)
- Dark Mode: Default (bg: #0F172A, card: #1E293B, text: #F1F5F9)

**\*\*Typography:\*\***

- Headings: Inter Bold (H1: 72px, H2: 48px, H3: 36px)
- Body: Inter Regular (18px)
- Code: JetBrains Mono

**\*\*Animations:\*\***

- Scroll-triggered fade-in + slide-up
- Number counters (0 â€™ final value)
- Progress bars fill on scroll
- Hover: Lift 4px + glow border

**\*\*Responsive:\*\***

- Desktop: 1440px+
- Laptop: 1024px
- Tablet: 768px
- Mobile: 375px

---

**## 15. ENGINEERING BEST PRACTICES**

**### Code Organization**

**\*\*MVC Separation:\*\***

- **\*\*Models:\*\*** Only database relationships + accessors/mutators
- **\*\*Controllers:\*\*** Only HTTP handling + validation + response
- **\*\*Services:\*\*** ALL business logic (Career Capital calculation, interview scoring, etc.)
- **\*\*Repositories:\*\*** Database queries (abstract SQL away from services)

**\*\*Example:\*\***



```

```php
// BAD (Business logic in controller):
public function update(Request $request) {
    $fellow = User::find($request->fellow_id);
    $fellow->score = $request->new_score;
    $fellow->tier = ($request->new_score >= 61) ? 'elite' : 'professional';
    $fellow->save();
}

// GOOD (Business logic in service):
public function update(UpdateCareerCapitalRequest $request) {
    $this->careerCapitalService->updateCareerCapital(
        fellowId: $request->fellow_id,
        trackId: $request->track_id,
        newScore: $request->new_score,
        adminId: auth()->id(),
        justification: $request->justification
    );
}
```

```

### ### SOLID Principles

#### \*\*S - Single Responsibility:\*\*

- Each class does ONE thing
- CareerCapitalService = Career Capital logic ONLY
- InterviewService = Interview logic ONLY

#### \*\*O - Open/Closed:\*\*

- Open for extension, closed for modification
- Use interfaces for flexibility
- Strategy pattern for different interview types

#### \*\*L - Liskov Substitution:\*\*

- Subclasses must be substitutable for parent classes
- Use abstract classes wisely

#### \*\*I - Interface Segregation:\*\*

- Many small interfaces > one large interface
- `Scoreable`, `Verifiable`, `Reportable`

#### \*\*D - Dependency Inversion:\*\*

- Depend on abstractions, not concretions
- Use dependency injection (Laravel Service Container)

### ### Security Best Practices

#### \*\*Input Validation:\*\*

- ALWAYS use Form Requests (never trust `\$request->input()` directly)
- Whitelist allowed values (enum validation)
- Type-cast inputs (integer, float, boolean)

#### \*\*SQL Injection Prevention:\*\*

- ALWAYS use Eloquent ORM (parameterized queries)
- NEVER use raw SQL with user input (unless absolutely necessary + parameterized)

#### \*\*XSS Prevention:\*\*

- Blade auto-escapes output: `{{ \$variable }}`
- If raw HTML needed, sanitize: `{!! clean(\$html) !!}`

#### \*\*CSRF Protection:\*\*

- Laravel automatically includes CSRF tokens in forms
- Always use `@csrf` directive in Blade forms

#### \*\*Authentication:\*\*

- Use Laravel Sanctum for API tokens
- bcrypt for password hashing (built-in)
- Rate limiting on login attempts (Laravel throttle middleware)

#### \*\*Authorization:\*\*

- Use Spatie Permission package for role-based access
- Check permissions in controllers: `auth()->user()->can('permission')`

#### \*\*Data Encryption:\*\*

- Sensitive data encrypted at rest: `encrypt(\$data)` / `decrypt(\$encryptedData)`
- HTTPS enforced in production

### ### Performance Optimization

#### \*\*Database Queries:\*\*

- ALWAYS use eager loading to prevent N+1 queries:  
```php  
// BAD:
\$fellows = Fellow::all();
foreach (\$fellows as \$fellow) {
 echo \$fellow->track->name; // N+1 query
}

// GOOD:
\$fellows = Fellow::with('track')->get();
foreach (\$fellows as \$fellow) {
 echo \$fellow->track->name; // No extra queries
}
```

#### \*\*Indexes:\*\*

- Add indexes on foreign keys
- Add indexes on frequently queried columns (`score`, `tier`, `status`)
- Composite indexes for multi-column queries

#### \*\*Caching:\*\*

- Cache admin settings (rarely change):  
```php  
\$eliteThreshold = Cache::remember('elite_tier_threshold', 3600, function() {
 return AdminSetting::where('key', 'elite_tier_threshold')->value('value');
});
```
- Cache recruiter profile view counts
- Cache Career Capital breakdowns (invalidate on score update)

#### \*\*Job Queues:\*\*

- AI interview processing (background job)
- Email sending (background job)
- Heavy calculations (background job)

#### \*\*Pagination:\*\*

- ALWAYS paginate large datasets (20-50 items/page)
- Use `simplePaginate()` when page numbers not needed (faster)

---

## ## 16. SECURITY & DATA PROTECTION

### ### Data Privacy

#### \*\*Sensitive Data:\*\*

- Passwords: bcrypt hashed (never store plain text)
- API keys: stored in `.env`, never in version control
- Personal info: encrypted if required (phone numbers, addresses)

#### \*\*GDPR/Privacy Compliance:\*\*

- Data export: Fellows can download all their data (JSON/PDF)
- Data deletion: Fellows can request account deletion (soft delete)
- Consent: Terms of Service + Privacy Policy acceptance on signup

### ### Access Control

#### \*\*Role-Based Access:\*\*

- Fellows: See only their own data
- Admins: See all data
- Recruiters: See only public/approved data

#### \*\*Row-Level Security:\*\*

- Query scoping:  
```php  
// Ensure fellow sees only their activities
Activity::where('fellow_id', auth()->id())->get();

// Global scope for soft deletes
Activity::withoutGlobalScope(SoftDeletingScope::class)->get();
```

### ### Audit Logging

**\*\*All sensitive actions logged:\*\***

- Career Capital score changes
- Track switches
- Activity approvals/rejections
- Recruiter intro requests
- Admin setting changes

**\*\*Audit Log Table:\*\***

- Immutable (never update/delete)
- Stores: Who, what, when, why
- Retention: Permanent (compliance)

---

## ## 17. QUALITY ASSURANCE REQUIREMENTS

### ### Testing Strategy

**\*\*Unit Tests (80%+ Coverage):\*\***

- Test all service layer methods
- Test repositories
- Test models (relationships, accessors, mutators)

**\*\*Feature Tests:\*\***

- Test complete user workflows:
  - Fellow submits activity â†’ Admin approves â†’ Score updates
  - Recruiter views profile â†’ Requests intro â†’ Admin approves
  - Fellow switches track â†’ Score recalculated

**\*\*Integration Tests:\*\***

- Test external API integrations (when added):
  - GitHub API
  - OpenAI API
  - Stripe API

**\*\*Browser Tests (Optional - Laravel Dusk):\*\***

- Test JavaScript-heavy features
- Test cross-browser compatibility

### ### Test Examples

```
```php
// Unit Test: CareerCapitalService
public function test_calculate_tier_returns_elite_for_score_above_threshold()
{
    $service = new CareerCapitalService();
    $tier = $service->calculateTier(87.5);
    $this->assertEquals('elite', $tier);
}

// Feature Test: Activity Submission
public function test_fellow_can_submit_activity()
{
    $fellow = User::factory()->create(['role' => 'fellow']);
    $track = Track::factory()->create();

    $response = $this->actingAs($fellow)->post('/fellow/activities', [
        'track_id' => $track->id,
        'type' => 'project',
        'title' => 'E-commerce Platform',
        'description' => 'Built with Laravel + React',
        'url' => 'https://github.com/akeem/ecommerce',
    ]);

    $response->assertStatus(302); // Redirect
    $this->assertDatabaseHas('activities', [
        'fellow_id' => $fellow->id,
        'title' => 'E-commerce Platform',
        'status' => 'pending',
    ]);
}
```
```

### ### Code Quality Checks

**\*\*Laravel Pint (PSR-12 Formatting):\*\***

```bash

```

./vendor/bin/pint
```

PHPStan (Static Analysis - Level 8):
```bash
./vendor/bin/phpstan analyse
```

PHP_CodeSniffer:
```bash
./vendor/bin/phpcs --standard=PSR12 app/
```

18. PERFORMANCE & OPTIMIZATION

Database Optimization

Indexes:
```sql
-- Foreign keys
INDEX idx_fellow_id ON activities(fellow_id)
INDEX idx_track_id ON activities(track_id)

-- Frequently queried columns
INDEX idx_score ON fellow_tracks(score)
INDEX idx_tier ON fellow_tracks(tier)
INDEX idx_status ON activities(status)

-- Composite indexes
INDEX idx_fellow_track ON fellow_tracks(fellow_id, track_id)
```

Query Optimization:

- Use `select()` to fetch only needed columns
- Use `limit()` for large datasets
- Avoid `SELECT *` (fetch specific columns)

Database Connection Pooling:

- Configure in `config/database.php`
- Max connections: 100 (adjust based on traffic)

Caching Strategy

Cache Drivers:

- Development: `file` or `array`
- Production: `redis` (fast, scalable)

What to Cache:

- Admin settings (1 hour TTL)
- Track lists (1 day TTL)
- Career Capital breakdowns (10 min TTL, invalidate on update)
- Public profiles (1 hour TTL)

Cache Tags (Redis):
```php
Cache::tags(['fellow', 'career-capital'])->put('fellow_87_cc', $data, 600);
Cache::tags(['fellow'])->flush(); // Clear all fellow-related cache
```

Frontend Optimization

Asset Optimization (Vite):

- Minification (automatic in production)
- Tree-shaking (removes unused JS/CSS)
- Code splitting (lazy load routes)

Image Optimization:

- Use WebP format (90% smaller than PNG)
- Lazy loading: `
- CDN for static assets (Cloudinary)

JavaScript:

- Minimize third-party libraries
- Defer non-critical JS: `

```

---

## ## 19. DEVELOPMENT WORKFLOW

### ### Git Workflow

#### \*\*Branching Strategy:\*\*

- `main` - Production-ready code (never commit directly)
- `develop` - Integration branch
- `feature/feature-name` - New features
- `bugfix/bug-description` - Bug fixes
- `hotfix/critical-issue` - Emergency fixes

#### \*\*Commit Messages (Conventional Commits):\*\*

```

```
feat: add Career Capital score calculation service
fix: resolve N+1 query in fellow dashboard
docs: update README with setup instructions
test: add unit tests for InterviewService
refactor: extract validation logic into FormRequest
```
```

#### \*\*Pull Request Process:\*\*

1. Create feature branch: `git checkout -b feature/multi-track-system`
2. Implement feature, write tests
3. Run tests: `php artisan test`
4. Run code quality checks: `./vendor/bin/pint && ./vendor/bin/phpstan analyse`
5. Commit with meaningful message
6. Push to GitHub: `git push origin feature/multi-track-system`
7. Create Pull Request (PR) with description
8. Code review by senior engineer
9. Merge to `develop` after approval
10. Deploy to staging for QA testing
11. Merge `develop` â†’ `main` for production deployment

### ### Environment Setup

#### \*\*Local Development:\*\*

```bash

1. Clone repository

```
git clone https://github.com/innova-cmr/iks-career-capital.git
cd iks-career-capital
```

2. Install dependencies

```
composer install
npm install
```

3. Environment configuration

```
cp .env.example .env
php artisan key:generate
```

4. Database setup

```
php artisan migrate
php artisan db:seed
```

5. Build assets

```
npm run dev
```

6. Start server

```
php artisan serve
```
```

#### \*\*Environment Variables (.env):\*\*

```

```
APP_NAME="IKS Career Capital"
APP_ENV=local
APP_DEBUG=true
APP_URL=http://localhost:8000
```

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=iks_career_capital
DB_USERNAME=root
DB_PASSWORD=
```

```
CACHE_DRIVER=file
QUEUE_CONNECTION=sync
SESSION_DRIVER=file
```

```
# OpenAI API (for interviews)
OPENAI_API_KEY=sk-...
```

```
# Stripe (for recruiter subscriptions)
STRIPE_KEY=pk_test_...
STRIPE_SECRET=sk_test_...
```
```

---

## ## 20. CRITICAL SUCCESS METRICS

### ### Platform Health Metrics

#### \*\*Fellows:\*\*

- Total enrolled: 500 (Year 1 target)
- Weekly active users: 40%+
- Tier distribution:
  - Rookie: 35%
  - Intern: 35%
  - Professional: 20%
  - Elite: 10%
- Average Career Capital score: 45%
- Churn rate: <5%

#### \*\*Interview Prep:\*\*

- Avg interviews per fellow: 25+
- Interview completion rate: 80%+
- Avg score: 75%+

#### \*\*Recruiters:\*\*

- Total subscribed: 50 (Year 1)
- Free tier: 30
- Partner tier: 15
- Premium tier: 5
- Monthly active recruiters: 70%+

#### \*\*Placements:\*\*

- Fellows hired: 100+ (Year 1)
- Avg time-to-hire: 12 days (vs industry 45 days)
- Avg salary: XAF 10M/year
- Fellow satisfaction: 4.5+ stars

#### \*\*Revenue:\*\*

- Year 1: XAF 24M (~\$40K USD)
- Year 3: XAF 105M (~\$175K USD)

### ### Technical Metrics

#### \*\*Performance:\*\*

- Page load time: <2 seconds
- Database query time: <100ms (avg)
- Uptime: 99.9%+
- API response time: <200ms

#### \*\*Code Quality:\*\*

- Test coverage: 80%+
- PHPStan level: 8
- Zero critical security vulnerabilities
- Code review approval rate: 95%+

---

## ## FINAL NOTES FOR ENGINEERS

### ### This Is a Flagship System

#### \*\*Treat it like enterprise software:\*\*

- Write code you'd be proud to show in 5 years
- Document everything (future developers will thank you)
- Test rigorously (bugs cost reputation)
- Optimize early (technical debt compounds)

### ### Quality Over Speed

**\*\*I-NNOVA CMR spent 3+ years building products before launch.\*\***

You have permission to:

- Refactor messy code
- Write comprehensive tests
- Request code reviews
- Push back on unrealistic deadlines

**\*\*Never compromise on:\*\***

- Security
- Data integrity
- User experience
- Code maintainability

### ### Ask Questions

**\*\*If anything in this spec is unclear:\*\***

- Ask the product team
- Discuss with fellow engineers
- Document your decisions (in code comments + README)

### ### Build for Africa

**\*\*This platform serves African talent.\*\***

Consider:

- Slow internet connections (optimize assets)
- Mobile-first design (70% of users on mobile)
- Data costs (minimize API calls, cache aggressively)
- Local payment methods (M-Pesa, Mobile Money - future)

...

## DOCUMENT END

```
Version: 1.0
```

**\*\*Last Updated:\*\*** February 2026

**\*\*Maintained by:\*\* I-NOVA CMR Engineering Team**

**\*\*Contact:\*\*** [engineering@innovacmr.com](mailto:engineering@innovacmr.com)

— — —

**\*\*This specification is a living document. Update it as the product evolves.\*\***

**\*\*Now go build something amazing.\*\*** ðŸš€ðŸ†˜ðŸ†˜²